SALESLENS ENTERPRISE BI PLATFORM • ACADEMIC DELIVERABLE DOC-04

Software Effort & Duration Estimation Sheet

Course: B.Tech CSE Semester III • Software Engineering & Project Management • Version 1.0

1. Primary Mathematical Formulas & Parameters

Total Effort Equation (E): $E = E_{\text{cleaning}} + E_{\text{views}} + E_{\text{testing}} + E_{\text{overhead}}$
Effective Daily Team Capacity (C): $C = \text{FTE}_{\text{BI}} (1.0) + \text{FTE}_{\text{Analyst}} (0.5) = \mathbf{1.50 \text{ Person-Days / Working Day}}$
Project Duration: $D = E / C$ (Days) • $W = D / 5$ (Weeks)
Feasibility Condition: Feasible if $W \le W_{\text{deadline}} = 8.00 \text{ Weeks}$ ($D \le 40.0 \text{ Working Days}$)

2. Comparative Mathematical Evaluation: 9-View vs. 26-Report Scope

Metric / Parameter	9-View In-Scope Dashboard (Chosen)	Full 26-Report Legacy Scope (Rejected)	Academic Rationale / Impact
Number of Views (N)	9 VIEWS	26 REPORTS	Scope pruned by 65% based on 80/20 rule
Effort per View (e_v)	3.0 Person-Days	3.0 Person-Days	Standardized UI charting templates
View Build Effort (E_v)	**27.0 Person-Days**	**78.0 Person-Days**	Saved 51.0 Person-Days of engineering
14% Data Cleaning (E_c)	8.0 Person-Days	8.0 Person-Days	Fixed complexity for 4-tier decision table
Verification & QA (E_t)	5.0 Person-Days	5.0 Person-Days	BVA, decision tables & DRE testing
TOTAL EFFORT (E)	40.0 Person-Days	91.0 Person-Days	**56.0% Overall Effort Reduction**
Working Days Duration (D)	**26.67 Working Days**	**60.67 Working Days**	Calculated as $E / 1.5 \text{ PD/day}$
Duration in Calendar Weeks (W)	**5.33 Weeks (≈ 5.4)**	**12.13 Weeks (≈ 12.2)**	Calculated as $D / 5 \text{ days/week}$
Target Deadline (W_{target})	8.00 Weeks (40 Days)	8.00 Weeks (40 Days)	Annual Sales Conference Milestone
Variance to Deadline ($Delta$)	+2.67 WEEKS BUFFER	-4.13 WEEKS OVERDUE	9-View solution leaves 33% safety buffer
Feasibility Status	PASSED (100% FEASIBLE)	FAILED (UNVIABLE)	Pruned scope guarantees conference success

3. Step-by-Step Intermediate Calculations

Scenario A: 9-View Baseline (Target Solution)

1. **View Effort:** $9 \text{ views} \times 3.0 \text{ PD} = 27.0 \text{ PD}$
2. **Total Effort:** $8.0 (\text{Clean}) + 27.0 (\text{Views}) + 5.0 (\text{QA}) = \mathbf{40.0 \text{ PD}}$
3. **Working Days:** $D = 40.0 / 1.5 = \mathbf{26.67 \text{ Days}}$
4. **Calendar Weeks:** $W = 26.67 / 5 = \mathbf{5.33 \text{ Weeks}}$
5. **Reserve Buffer:** $8.00 - 5.33 = \mathbf{+2.67 \text{ Weeks (13.3 Days)}}$

✓ Healthy 33% buffer absorbs edge-case anomalies.

Scenario B: 26-Report Legacy (Why it was Rejected)

1. **View Effort:** $26 \text{ reports} \times 3.0 \text{ PD} = 78.0 \text{ PD}$
2. **Total Effort:** $8.0 (\text{Clean}) + 78.0 (\text{Views}) + 5.0 (\text{QA}) = \mathbf{91.0 \text{ PD}}$
3. **Working Days:** $D = 91.0 / 1.5 = \mathbf{60.67 \text{ Days}}$
4. **Calendar Weeks:** $W = 60.67 / 5 = \mathbf{12.13 \text{ Weeks}}$
5. **Schedule Deficit:** $8.00 - 12.13 = \mathbf{-4.13 \text{ Weeks (Overdue by 21 Days)}}$

✕ Misses the annual sales conference by a month.

4. Software Engineering Principles in Estimation

- **Cone of Uncertainty (Boehm & McConnell):** At inception, estimation variance spans $0.25\times$ to $4.0\times$. Freezing the 9 views in Week 1 and locking the 4-tier decision table narrows uncertainty to $\pm 10\%$.
- **Brooks' Law (The Mythical Man-Month):** *"Adding manpower to a late software project makes it later."* Introducing additional personnel midway adds communication overhead without speeding up sequential ETL data cleaning logic.
- **Parkinson's Law:** Work expands to fill available time. We counteract this through weekly working prototype demos with the Sales Director every Friday.
- **Murphy's Law in Legacy Pipelines:** The 2.67-week buffer explicitly protects against unexpected legacy data quirks (e.g. malformed postal codes, foreign customer addresses).